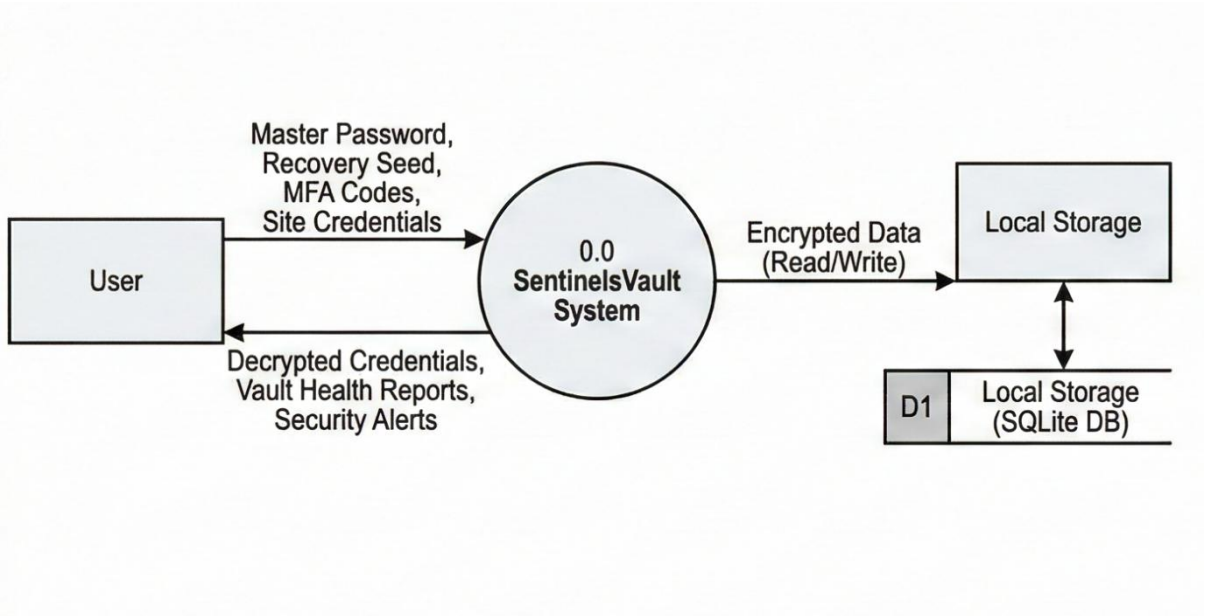
Data Flow Diagram

Level 0 DFD: Context Diagram

The Level 0 DFD provides a bird's-eye view. The system is treated as a single process interacting with the User and the Local Storage.

- **User Inputs:** Master Password, Recovery Seed, MFA Codes, and Site Credentials.
- **System Outputs:** Decrypted Credentials, Vault Health Reports, and Security Alerts.
- **Data Store:** The SQLite Database (Encrypted at Rest).

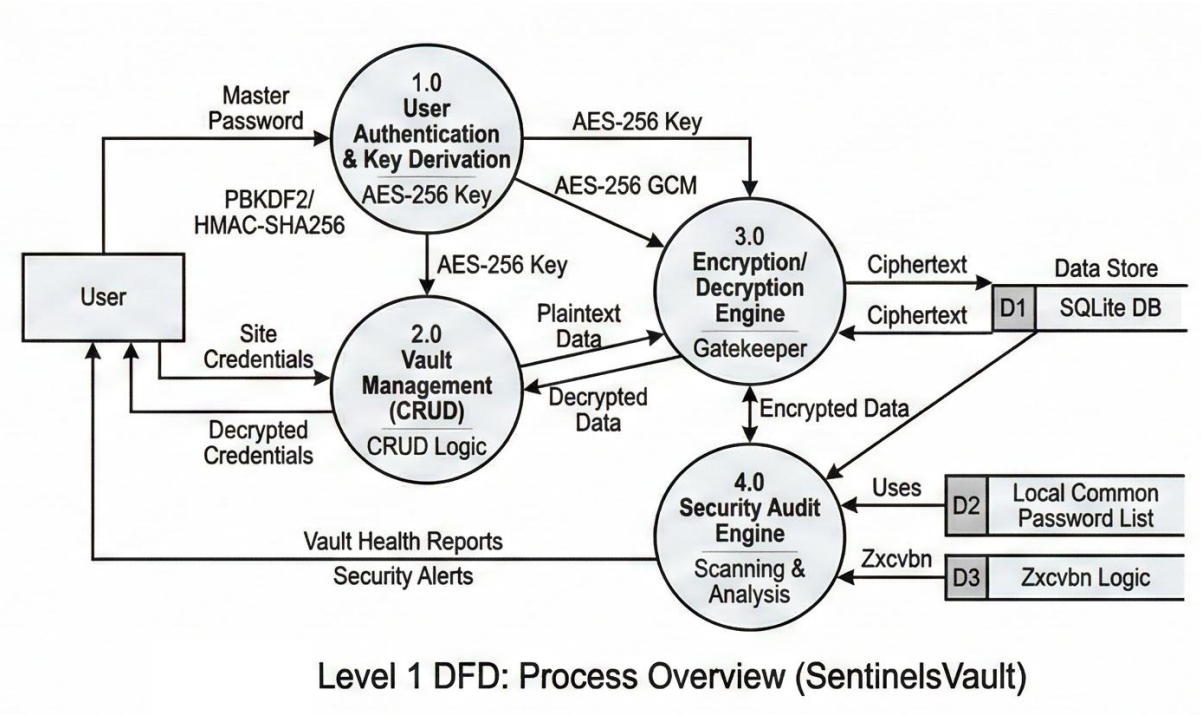


Level 1 DFD: Process Overview

Level 1 breaks the system into its core functional modules: Authentication, Credential Management, and the Security Auditing Engine.

Key Processes:

- **P1: User Authentication & Key Derivation:** Processes the Master Password through PBKDF2/HMAC-SHA256 to create the AES-256 key.
- **P2: Vault Management (CRUD):** Handles the input, update, and deletion of encrypted credentials.
- **P3: Encryption/Decryption Engine:** The gatekeeper between the UI and the Database, ensuring data is never stored in plaintext.
- **P4: Security Audit Engine:** Scans the database for weak, reused, or leaked passwords.



Level 1 DFD: Process Overview (SentinelsVault)

Level 2 DFD: Detailed Encryption & Audit Logic

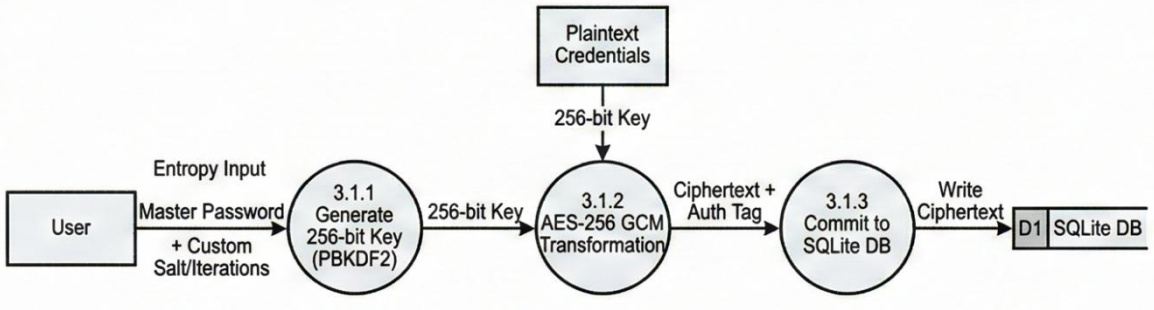
This level focuses on the "Zero-Knowledge" aspect and the specific data flow during a security audit.

Encryption Flow

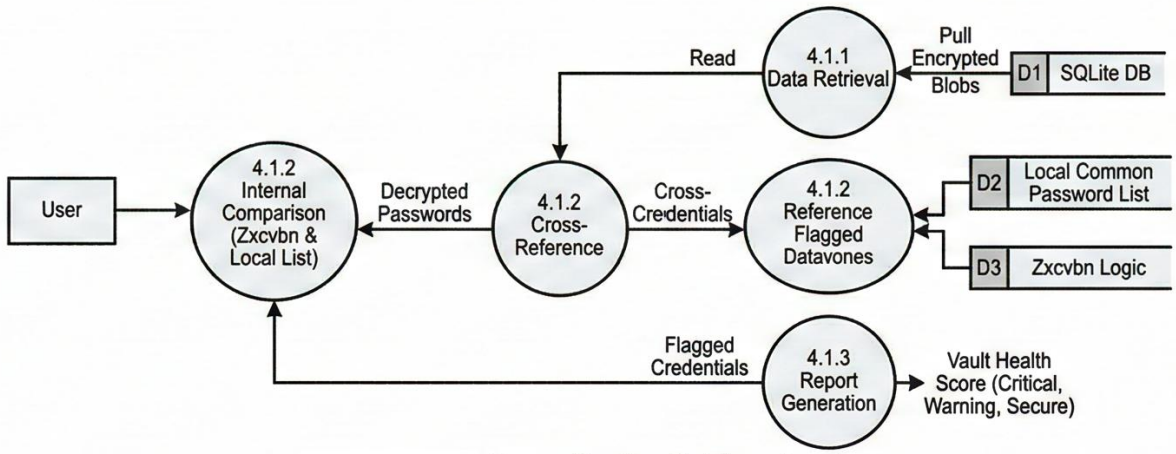
1. **Entropy Input:** User provides Master Password + Custom Salt/Iterations.
2. **Key Generation:** PBKDF2 outputs a 256-bit key.
3. **GCM Transformation:** Plaintext credentials + 256-bit key $\rightarrow$ Ciphertext + Auth Tag.
4. **Storage:** Ciphertext is committed to the ACID-compliant SQLite DB.

Security Audit Flow

1. **Data Retrieval:** The system pulls encrypted blobs.
2. **Internal Comparison:** Passwords are cross-referenced against the **Local Common Password List** (Top 10,000) and **Zxcvbn logic**.
3. **Report Generation:** Data is aggregated into a **Vault Health Score** (Critical, Warning, or Secure).



Encryption Flow Logic



Security Audit Flow Logic